

SHUHAN (Alice) AI

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EDUCATION

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- PhD, School of Education, University of California, Los Angeles, CA** 2023.09 – 2027.06 (expected)
- Major: Higher Education & Organizational Change (HEOC); Candidacy on December 19, 2025
 - Dissertation Committee: Kevin Eagan (Chair); Jordan Rickles (Co-Chair); Cecilia Rios-Aguilar; Linda J. Sax
- M.S., Department of Statistics, University of California, Los Angeles, CA** 2025.09 – 2027.06 (expected)
- Major: Statistics and Data Science; Advisor: Qing Zhou
- M.S. Ed, University of Pennsylvania, Philadelphia, PA** 2021.09 – 2022.12
- Major: Higher Education, Graduate School of Education
- BA, Northeast Normal University, Jilin, China** 2017.09 – 2021.06
- Major: Education, School of Education Science

ACADEMIC APPOINTMENTS

Teaching Assistant

- EDUC230A Introduction to Research Design and Statistics (graduate course) 2025.09 – 2026.03
- EDUC230B: Linear Statistical Models: Analysis of Designed Experiments (graduate course) 2026.03 – 2026.06

Graduate Student Researcher (PI: Dr. Jordan Rickles), UCLA, CA

School of Education and Information Studies

2026.03 – 2026.06

Project: National Trends and Prevalence of Credit Recovery in Secondary Education

- Construct school-level analysis files by merging multi-source, including the U.S. Department of Education Civil Rights Data Collection (CRDC), Common Core Data (CCD), and Stanford Education Data Archive (SEDA).
- Conduct descriptive and exploratory analyses of credit recovery prevalence and access disparities across U.S. public high schools, establishing an empirical foundation for equity-focused secondary education policy.

Graduate Student Researcher (PI: Dr. Kevin Eagan), UCLA, CA

School of Education and Information Studies

2023.09 – 2025.06

Project: The Effects of NIH-BUILD Participation on the Development of Science Identity and STEM Aspirations

- Employed quasi-experimental methods (propensity score modeling, IPW) to evaluate the causal impact of the NIH-BUILD initiative on underrepresented students' science identity, STEM graduate degree aspirations.
- Delivered actionable insights to inform policies on mentoring, research opportunities, and financial support to enhance student persistence and STEM degree attainment.

Journal Editor, UCLA, CA

InterActions: UCLA Journal of Education and Information Studies

2024.01 – 2025.06

- Coordinated and managed journal operations, communicating with the Journal Advisory Board and Research Groups.
- Supervised manuscript submission, peer-review processes, and editorial workflows to support authors and reviewers.
- Maintained and updated the journal's eScholarship platform, ensuring accuracy for all published content.

AWARDS & FELLOWSHIPS & FUNDINGS

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- Society for Research on Educational Effectiveness (SREE) Summer Research Fellowship, \$8,000 2026
 - UCLA Dissertation Year Fellowship, \$30,000/year 2026, 2027
 - UCLA William and Evelyn Scholarship, \$30,000/year 2023, 2025
 - UCLA Division of Graduate Education Fellowship, 15,000/year 2024, 2025
 - UCLA Graduate Summer Research Mentorship Award (GSRM), \$6,000/year 2024, 2025
 - ASHE Graduate Student Travel Scholarship 2025
 - Excellent Student, The First Prize Scholarship 2018, 2019, 2020
 - Outstanding Graduates Award, Social Practice Award 2021
 - Fundamental Research Award for the Central Universities 2018
 - Scholarship for Exchange Student 2018

RESEARCH INTERESTS

STEM; Higher Education; AI in Education; Causal Inference; Network Analysis; Statistics

PUBLICATIONS

Refereed Journal Articles

- Ai, S.** & Eagan, K., (2026). From Club to College: The Impact of Science Clubs/Groups on High School Students' STEM Major Choice. *Research in Higher Education*, 67, 32. <https://doi.org/10.1007/s11162-026-09892-8>
- Ai, S.** & Eagan, K., Wu, J. (2026). Fueling Ambition: Graduate Degree Aspirations Among Women of Color in STEM. *CBE – Life Science Education*. <https://doi.org/10.1187/cbe.25-10-0229>
- Ai, S.** (2024). Empowering Online Teaching: A System Review of Online Instructors' Professional Development in Higher Education. *InterActions: UCLA Journal of Education and Information Studies*, 19(1). <https://doi.org/10.5070/D41.6661>

Research Preprints

- Ai, S.** (2026). Causal Fairness Analysis of ADHD Status and High School STEM Outcomes. *arXiv preprint*. Retrieved from <https://arxiv.org/abs/2604.10824>
- Ai, S.** (2026). Discussion Network Formation and Evolution in an Online Professional Development Class: Evidence from a MOOC for K-12 Educators. *arXiv preprint*. Retrieved from <https://arxiv.org/abs/2601.01117>
- Ai, S.**, Kevin, E., Olivares, J., & Jiang, T. (2024). Can Community Cultural Wealth Maintain STEM Career Aspirations of Undergraduate Women of Color? Examining HERI Data. In annual meeting of the *Association for the Study of Higher Education (ASHE)*, Minneapolis, MN. Retrieved from <https://escholarship.org/uc/item/86c6r5m2>

Technical Reports

- Omery, F., Montes-Diaz, I., **Ai, S.** (2023). City Fiscal Conditions. National League of Cities. Retrieved from <https://www.nlc.org/resource/city-fiscal-conditions-2023/>
- Omeyr, F. & **Ai, S.** (2023). State of the Cities. National League of Cities. Retrieved from <https://www.nlc.org/resource/state-of-the-cities-2023/>

Under Peer-Review

- Ai, S.** (R & R). Engaging Future Scientists: The Effect of Science Museum Visits on High School Students' Science Identity Development. *International Journal of STEM Education*.
- Ai, S.** (under review). Building Professional Ties Online: How Discussion Networks Form and Evolve in a MOOC for K-12 Educators. *Social Networks*.
- Ai, S.** & Eagan, K., (under review). Can Research Mentorship Fuel Undergraduate Women of Color's Aspirations Pursuing a STEM Graduate Degree? Examining HERI Data. *American Educational Research Journal*.
- Ai, S.** & Wu, J. (under review). Shortcut or Stepping-Stone? AI-Assisted Coursework and STEM Graduate Aspirations among U.S. STEM Undergraduates. *AERA Open*.
- Wu, J., Li, L., & **Ai, S.** (invited submission, 7% acceptance rate). Finding the Way In: Longitudinal Predictors of Female STEM Joiner Pathways. *AERA Open*. Special Topic Issue: *Engaging and Retaining Individuals in STEM*
- Eagan, M. K., Ramos, H., **Ai, S.**, Romero, A. L., & Cobian, K. (under review). BUILDing science identity: Exploring the efficacy and moderation effects of a nationally funded intervention. *Educational Evaluation and Policy Analysis*.
- Jiang, T., **Ai, S.**, Jiao, C., Li, K. (under review). Revealing Hidden Contributions on Research Data: The Potential of Data Paper. *Scientometrics*.

SELECTED TUTORIAL POSTS

- Ai, S.** (2025–2026). [Causal Inference Tutorial Series](#): Potential Outcomes, Matching, Weighting, DID/RDD, DAGs/SCM, Causal Fairness Analysis.
- Ai, S.** (2025–2026). [Social Network Analysis R Tutorial Series](#): Basic Visualization, ERGM, TERGM.
- Ai, S.** (2025–2026). [Machine Learning Methods Series](#): Regression, Gradient Descent, Transformers, Diffusion Models, Practical ML Guide in R.
- Ai, S.** (2026). [Bayesian Methods Tutorial Series](#): Introduction to Bayesian, Specifying Prior and Posterior Inference, Regression with `rstanarm`.

CONFERENCE PAPER PRESENTATION

- Eagan, K., & **Ai, S.** (2026). BUILDing STEM Futures: The Role of Undergraduate Research and Mentorship in Developing Graduate Degree Aspirations. Paper presentation at the annual meeting of the American Education Research Association, Division J: Postsecondary Education, Los Angeles, CA.
- Ai, S.** (2026). Does Exposure to Science Museums or Planetariums Shape Science Identity and STEM Major Choice? Evidence from HSLs:09. Roundtable presentation at the annual meeting of the American Education Research Association, Division J: Postsecondary Education, Los Angeles, CA.
- Ai, S.,** Eagan, K., & Wu, J. (2025). Fueling Ambition: Graduate Degree Aspirations Among Women of Color in STEM. Paper presentation at the annual meeting of the Association for the Study of Higher Education, Denver, CO.
- Ai, S.,** Eagan, K. (2025). What Factors Promote High School Students' Development of Science Identity and STEM Major Aspiration? An Analysis of HSLs:09 Data. Roundtable presentation at the annual meeting of the American Education Research Association, Division J: Postsecondary Education, Denver, CO.
- Ai, S.** (2025). A Critical Race Analysis of Black and Hispanic Female Graduate Students Enrollment Trends from 2012 to 2021. Roundtable presentation at the annual meeting of the American Education Research Association, Division J: Postsecondary Education, Denver, CO.
- Ai, S.,** Rios, M., Olivares, J., & Jiang, T. (2024). Can Community Cultural Wealth Guide Undergraduate Women of Color Toward STEM Careers? Examining HERI Data. Paper presentation at the annual meeting of the Association for the Study of Higher Education, Minneapolis, MN.
- Ramos, H., Eagan, K., & **Ai, S.** (2024). BUILDing Science Identity: Exploring the Efficacy and Moderation Effects of a Nationally Funded Intervention. Paper presentation at the annual meeting of the Association for the Study of Higher Education, Minneapolis, MN.
- Jiang, T., **Ai, S.,** & Li, K. (2024). Revealing Contributions on Research Data: The Potential of Data Paper. Paper presentation at the MET/STI 2024 Workshop of the Association for Information Science and Technology (Virtual).
- Ai, S.** (2024). STEM Curriculum as a Catalyst: Narrowing Racial and Structural Inequalities. Roundtable presentation at the annual meeting of the American Education Research Association, Division L: Educational Policies and Politics, Philadelphia, PA.
- Liu, Xi., & **Ai, S.** (2024). Improving Affect Detections for K-12 Online Math Learning Using Time-related Variables. Paper presentation at the annual meeting of the American Education Research Association, Division C: Learning and Instruction, Philadelphia, PA.

INVITED TALKS

- Ai, S.** (2026). From Club to College: The Impact of Science Clubs/Groups on High School Students' STEM Major Choice. Invited talk at University of California, Los Angeles, CA.

PARTICIPATION IN FUNDED GRANTS

UCLA Dissertation Year Fellowship <i>Principal Investigator</i> <i>The Role of Community Service in Fostering Commitment to Social Responsibility Among STEM Students</i> Funded by UCLA Graduate Education Program: \$3,0000	2026.09-2027.07
Society for Research on Educational Effectiveness (SREE) & Harvard University CEPR <i>Summer Research Fellow; Principal Investigator</i> <i>Middle School Predictors of College and Career Success</i> Funded by SREE Summer Fellows Program: \$8,000	2026.06-2026.09
National Institute of General Medical Sciences (Grant No. U54GM119024) <i>Graduate Student Researcher</i> <i>Diversity Program Consortium Coordination and Evaluation Center at UCLA</i> Funded by National Institutes of Health (NIH)	2023.09-2025.06 2015-2025
UCLA Graduate Summer Research Mentorship Award (GSRM 2025) <i>Principal Investigator</i> <i>Race, Identity, and Context: A Multilevel Modeling Analysis of Senior Women's STEM Graduate Degree Aspirations</i> Funded by UCLA Fellowship: \$6,000	2025.06-2025.09
UCLA Graduate Summer Research Mentorship Award (GSRM 2024) <i>Principal Investigator</i> <i>Factors Promote High School Students' Science Identity and STEM Major Aspiration? Evidence from HSLS:09</i> Funded by UCLA Fellowship: \$6,000	2024.06-2024.09

COMMUNITY SERVICE

Reviewer for <i>AERA Open</i>	2026.05
Reviewer for <i>Journal of Women and Minorities in Science and Engineering</i>	2025.04
Reviewer & Chair for Annual Meeting of the Association for the Study of Higher Education	2024-2026
Reviewer & Chair & Discussant for Annual Meeting of the American Education Research Association	2023-2026

RECENT RESEARCH PROJECTS (PI)

Can Community Service Foster Commitment to Social Responsibility Among First-Year STEM Major Aspirants? <i>PhD Dissertation, UCLA School of Education and Information Studies</i> 2025.06 – present	
- Conducting a multilevel quasi-experimental study examining how community service participation shapes socially responsible outcomes (civic engagement, social agency, community commitment) among STEM undergraduates. - Analyzing HERI longitudinal data (Freshman Survey and College Senior Survey) and U.S. Integrated Postsecondary Education Data System (IPEDS) data, linking first-year experiences to senior-year outcomes. - Integrating undergraduate socialization theory with expectancy-value theory to explain how community-based interventions broaden the social impact of STEM education. - Proposing community-based solutions to empower STEM higher education in the era of AI, offering policy recommendations for cultivating socially responsible technology professionals through civic-oriented practice.	
Empirical Implications of Causal Inference: A Comparison of the Potential Outcomes Framework and Structural Causal Model <i>MS Statistics Thesis, UCLA School of Statistics & Data Science</i> 2025.10 – present	
- Conducting a systematic methodological comparison of the two dominant causal inference paradigms (Rubin's potential outcomes framework and Pearl's structural causal model), examining their empirical implications, assumptions, and practical trade-offs for applied social science research. - Evaluating identification strategies, sensitivity analyses, and estimand interpretations across both frameworks using simulated and empirical educational datasets. - Developing applied guidance for researchers using causal inference to address selection bias, unmeasured confounding, and mediation in observational studies, bridging statistical theory and applied social science practice.	

Discussion Network Formation and Evolution in an Online Professional Development Class: Evidence from a MOOC for K-12 Educators

Principle Investigator; Working Paper

2025.09 – present

- Analyzed peer discussion network formation among K-12 educators in a MOOC-Ed, applying cross-sectional and temporal exponential random graph models (ERGMs and TERGMs) to identify mechanisms driving interactions.
- Revealed that network activity peaked mid-course before concentrating among a tightly connected core, offering design recommendations for sustaining educator engagement in online professional development.
- Demonstrated proficiency in advanced network analysis methods (ERGM, TERGM, network visualization) applied to real-world educational data, with implications for both online education and professional development policy.

From Club to College: The Impact of Science Clubs/Groups on High School Students' STEM Major Choice

Principal Investigator; Paper in Press & Working Paper

2025.05 – present

- Investigated whether participation in high school science clubs/groups influences students' science identity development and subsequent STEM major choice using nationally representative federal HSLs:09 data (N = 23,503).
- Applied quasi-experimental design to assess how extracurricular science engagement shapes STEM major aspirations.
- Employed causal tree methods to detect heterogeneous treatment effects, identifying subgroups for whom science club participation had the strongest impact on STEM major choice across demographic and socioeconomic characteristics.
- Offered policy-relevant recommendations for K-12 schools and out-of-school programs seeking to broaden the STEM pipeline through accessible, community-based extracurricular interventions.

WORK EXPERIENCES

[National League of Cities](#), Washington, D.C.

Research and Data Consultant, Center for Research and Data Analysis

2023.01 – 2023.09

- Co-authored two nationally cited policy reports, State of the Cities and City Fiscal Conditions, synthesizing survey data from hundreds of municipalities to inform federal and local fiscal policy discussions.
- Developed natural language processing pipelines in Python and STATA to automate the classification of American Rescue Plan Act (ARPA) spending categories, reducing manual coding time and improving coding consistency.
- Cleaned, merged, and validated large-scale municipal datasets for the Preemption Project, supporting research on state-level legislative constraints on local governance.

[Wharton Online](#), Philadelphia, PA

Management Assistant, Wharton Executive Education

2021.09 – 2023.06

- Produced weekly analytics reports on learner engagement and course performance across 30+ MOOCs serving 2.3 million learners, informing strategic decisions on course design and platform optimization.
- Partnered with Coursera, Emeritus, and Canvas teams to evaluate and refine online course structures, contributing to improved learner retention for B2B executive education programs.
- Supported the development and launch of new B2B specializations, conducting course design evaluations and analyzing student interaction data to enhance learning outcomes.

COLLEGE ACTIVITIES

Summer Institutes in Computational Social Science (SICSS-UCLA), Los Angeles, CA

Participant

2024.06 – 2024.07

- Gained comprehensive training in advanced computational methods and their applications in social science research.
- Collaborated on interdisciplinary projects, employing techniques such as network analysis, and machine learning.
- Enhanced proficiency in utilizing statistical software and programming languages for data analysis, contributing to innovative solutions in higher education research.
- Applied the Potential Outcome Model for causal inference, developing skills in estimating treatment effects and addressing confounding factors.

The Walnut Street Club, University of Pennsylvania, GSE, Philadelphia, PA

Co-founder

2022.05 – 2023.06

- Co-designed and delivered an eight-week online research methods boot camp reaching over 200 graduate students, covering topics in STEM education, social-emotional learning, and education policy research.
- Established partnerships with Penn Catalyst and Montessori Charter School to bridge educational research and practice, facilitating knowledge exchange between researchers and K-12 practitioners.

- Curated and disseminated a biweekly research digest synthesizing recent publications from leading journals (e.g., Higher Education Policy, American Journal of Education), building a community of 200+ engaged readers.

Penn China Education Summit (PCES), Philadelphia, PA

Education Technology Panel Director (20h/week)

2021.12 – 2022.05

- Recruited and coordinated a panel of 20+ international scholars and 20+ education technology practitioners from China, the U.S., and the U.K., directing content strategy for the summit's largest thematic track.
- Organized 2 forums and 2 roundtable discussions that attracted 860 registrations and 1,380 total viewers, establishing PCES as a leading cross-national platform for education technology dialogue.
- Produced multimedia content including education podcasts and video recordings, extending the summit's reach.

PERSONAL SKILLS

Statistical Analyses	R; Python; STATA; SPSS; C++; Mplus; FlexMIRT; RapidMiner; Qualtrics; Latex
Statistical Methods	Linear Regression; Causal Inference; Social Network Analysis, Machine Learning
Languages	English (professional proficiency); Mandarin (native)
Sports	National Collegiate Table Tennis Association Member; Tennis; Swimming
Arts	Chinese Traditional Calligraphy; Oil Painting

REFERENCES

Associate Professor, Kevin Eagan

School of Education and Information Studies

Email: keagan@ucla.edu

Assistant Professor, Jordan Rickles

School of Education and Information Studies

Email: rickles@seis.ucla.edu

Professor, Manuel S. Gonzalez Canche

Graduate School of Education, University of Pennsylvania

Email: msgc@upenn.edu

Professor, Qing Zhou

Department of Statistics & Data Science

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